

THE CHOW RANK OF THE FOUR-BY-FOUR PERMANENT

ABSTRACT. We prove over every field of characteristic zero that

$$\text{ChowRank}(\text{perm}_4) = 8.$$

The lower bound uses the second-derivative space, a classification of its low-rank quadrics, a two-quotient rank inequality, and an exact rational chart minor. The upper bound is Glynn's eight-term decomposition. Independent programs reconstruct every computation used in the proof.

1. THE FOUR-BY-FOUR PERMANENT

1.1. **Definitions and statement.** Work over $\mathbb{C}$, and let

$$V = \text{span}\{x_{ij} : 0 \leq i, j \leq 3\}, \quad \dim V = 16.$$

Definition 1.1. The Chow rank of $f \in \text{Sym}^4 V$ is the least integer r for which

$$f = \sum_{s=1}^r \ell_{s1} \ell_{s2} \ell_{s3} \ell_{s4}, \quad \ell_{st} \in V.$$

Each product $\ell_1 \ell_2 \ell_3 \ell_4$ is a Chow term.

Theorem 1.2.

$$\boxed{\text{ChowRank}(\text{perm}_4) = 8}.$$

Section 1.8 gives an eight-term decomposition. The preceding sections prove that no decomposition with seven terms exists.

1.2. **Second derivatives and the Koszul flattening.** For $f \in \text{Sym}^4 V$, let

$$C_{2,2}(f) : \text{Sym}^2 V^* \longrightarrow \text{Sym}^2 V, \quad D \longmapsto D(f),$$

and set $E_f := \text{im } C_{2,2}(f)$. Define

$$\delta : \text{Sym}^2 V \otimes V \longrightarrow V \otimes \bigwedge^2 V$$

by

$$\delta((uv) \otimes w) = v \otimes (u \wedge w) + u \otimes (v \wedge w), \tag{1}$$

extended linearly. We use the flattening

$$K(f) := \delta \circ (C_{2,2}(f) \otimes \text{id}_V),$$

for which

$$\text{im } K(f) = \delta(E_f \otimes V). \tag{2}$$

In monomial bases the middle catalecticant is symmetric. Transposing gives

$$\text{row } K(f) \subseteq E_f \otimes V^*. \tag{3}$$

Write $E := E_{\text{perm}_4}$. Direct differentiation gives

$$E = \text{span}\{x_{ia}x_{jb} + x_{ib}x_{ja} : 0 \leq i < j \leq 3, 0 \leq a < b \leq 3\}. \tag{4}$$

These quadrics have distinct row-pair and column-pair supports, so $\dim E = \binom{4}{2}^2 = 36$.

Proposition 1.3.

$$\text{rank } K(\text{perm}_4) = 560.$$

For every Chow term $T = \ell_1 \ell_2 \ell_3 \ell_4$,

$$\text{rank } K(T) \leq 92,$$

with equality when $\ell_1, \dots, \ell_4$ are linearly independent.

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Proof. The domain of $\delta|_{E \otimes V}$ has dimension $36 \cdot 16 = 576$. If all first derivatives of $g \in \text{Sym}^3 V$ lie in E , define

$$\iota(g) := \sum_{a=0}^{15} \frac{\partial g}{\partial x_a} \otimes x_a.$$

Equation (1) gives $\delta(\iota(g)) = 0$, while tensor multiplication sends $\iota(g)$ to $3g$ by Euler's identity. Thus ι is injective.

For every three-row set $I \subset \{0, 1, 2, 3\}$ and three-column set $J \subset \{0, 1, 2, 3\}$, let

$$p_{I,J} := \text{perm}(x_{ij})_{i \in I, j \in J}.$$

The sixteen such cubics have disjoint supports, and all their first derivatives lie in E . They therefore give sixteen independent kernel vectors. Hence $\text{rank } K(\text{perm}_4) \leq 576 - 16 = 560$. In the bases from (4), exact elimination finds a 560×560 integer minor that is nonzero modulo 1000003. The same determinant is a nonzero integer, so the reverse inequality holds in characteristic zero.

For a term with independent factors, a $GL(V)$ -change of coordinates gives $T = x_0 x_1 x_2 x_3$. Then

$$E_T = \text{span}\{x_i x_j : 0 \leq i < j \leq 3\}, \quad \dim E_T = 6.$$

The four squarefree cubics in x_0, x_1, x_2, x_3 give four independent kernel vectors, so $\text{rank } K(T) \leq 96 - 4 = 92$. Exact elimination finds a nonzero 92×92 minor modulo 1000003, giving equality. Independent factor quadruples form a dense open set. Since the locus where matrix rank exceeds 92 is open, no degenerate quadruple can have larger rank: such an open set would meet the dense independent locus. Thus every Chow term has rank at most 92. $\square$

Remark 1.4. The modular calculations certify nonzero integer minors; they are not probabilistic rank estimates. Section 1.9 gives the exact reconstruction commands.

1.3. Low-rank quadrics in the derivative space. Index the basis in (4) by the six row pairs and six column pairs. A quadric $q \in E$ corresponds to a 6×6 coefficient matrix C . For a fixed row pair $i < j$, write the corresponding row as a symmetric zero-diagonal 4×4 matrix

$$D_{ij}[a, b] = C[(i, j), (a, b)] \quad (a \neq b), \quad D_{ij}[a, a] = 0.$$

Grouping the variables by matrix rows, the symmetric coefficient matrix is

$$M(q) = (D_{ij})_{0 \leq i, j \leq 3}, \quad D_{ji} = D_{ij}, \quad D_{ii} = 0. \quad (5)$$

Using the Hessian instead changes this matrix only by a nonzero scalar.

Lemma 1.5. *Let $D \neq 0$ be a symmetric zero-diagonal matrix over $\mathbb{C}$.*

- (i) $\text{rank } D \neq 1$.
- (ii) *If $\text{rank } D = 2$, and H is symmetric and zero-diagonal with $\text{im } H \subseteq \text{im } D$, then H is a scalar multiple of D .*

Proof. A rank-one symmetric matrix has the form $\lambda u u^\top$. A zero diagonal gives $\lambda u_i^2 = 0$ for every i , contradicting $D \neq 0$.

If $\text{rank } D = 2$, choose independent a, b such that

$$D = ab^\top + ba^\top.$$

The zero diagonal gives $a_k b_k = 0$ for every k , so the supports of a and b are disjoint. Symmetry and $\text{im } H \subseteq \text{span}(a, b)$ give

$$H = \begin{bmatrix} a & b \end{bmatrix} R \begin{bmatrix} a & b \end{bmatrix}^\top$$

for a symmetric 2×2 matrix R . The diagonal entries on the nonempty supports of a and b force $R_{11} = R_{22} = 0$, leaving only one off-diagonal parameter. Hence H is a scalar multiple of D . $\square$

Theorem 1.6. *If $0 \neq q \in E$ and $\text{rank } M(q) \leq 4$, then*

$$M(q) = U \otimes W,$$

where U and W are symmetric zero-diagonal 4×4 matrices of rank 2. In particular, $\text{rank } M(q) = 4$.

Proof. Choose a nonzero block $D := D_{ij}$. The associated principal submatrix is

$$N = \begin{bmatrix} 0 & D \\ D & 0 \end{bmatrix}, \quad \text{rank } N = 2 \text{rank } D \leq \text{rank } M(q) \leq 4.$$

Lemma 1.5 implies $\text{rank } D = 2$, hence $\text{rank } N = 4 = \text{rank } M(q)$. Thus all remaining block columns lie in the column space of N . For each k ,

$$\text{im } D_{ik}, \text{ im } D_{jk} \subseteq \text{im } D,$$

so Lemma 1.5(ii) gives scalars a_k, b_k with

$$D_{ik} = a_k D, \quad D_{jk} = b_k D.$$

Let G be a generalized inverse satisfying $DGD = D$, and put

$$N^- = \begin{bmatrix} 0 & G \\ G & 0 \end{bmatrix}.$$

Because the full matrix and N have the same rank, the generalized Schur complement vanishes. Therefore

$$\begin{aligned} D_{kl} &= \begin{bmatrix} D_{ik} \\ D_{jk} \end{bmatrix}^\top N^- \begin{bmatrix} D_{il} \\ D_{jl} \end{bmatrix} \\ &= (a_k b_l + b_k a_l) D. \end{aligned}$$

All blocks are scalar multiples of D , so $M(q) = U \otimes W$ with $W = D$ and U symmetric and zero-diagonal. Finally,

$$4 = \text{rank } M(q) = \text{rank } U \cdot \text{rank } W.$$

Neither nonzero zero-diagonal factor can have rank one, so both ranks are two. $\square$

Corollary 1.7. *For every subspace $L \subseteq V$ with $\dim L \leq 4$,*

$$\dim(E \cap \text{Sym}^2 L) \leq 1.$$

Consequently, if $F = E_T$ for a Chow term T , then

$$\dim(E \cap F) \leq 1.$$

Proof. If $0 \neq q \in E \cap \text{Sym}^2 L$, then $\text{rank } M(q) \leq \dim L \leq 4$. Theorem 1.6 shows that the rank is four and that the essential variable space is $\text{im } U \otimes \text{im } W$, also of dimension four. Thus $\dim L = 4$ and L equals this essential space.

Tensor-product factors of nonzero two-dimensional subspaces are unique: if $S_1 \otimes T_1 = S_2 \otimes T_2$, then $S_1 = S_2$ and $T_1 = T_2$. Within a fixed two-dimensional image, Lemma 1.5(ii) makes the rank-two symmetric zero-diagonal matrix unique up to scale. Hence any two nonzero elements of $E \cap \text{Sym}^2 L$ are proportional.

For a Chow term, every second derivative uses only $L = \text{span}(\ell_1, \dots, \ell_4)$, so $F \subseteq \text{Sym}^2 L$. The second claim follows, including repeated or dependent factors. $\square$

1.4. The chart calculation. For $0 \neq v \in V$, define

$$\psi_v : \text{Sym}^2 V/E \longrightarrow \text{coker}(\delta(E \otimes V)), \quad [q] \longmapsto [\delta(q \otimes v)]. \quad (6)$$

Its domain has dimension $136 - 36 = 100$. Since $\delta(v^2 \otimes v) = 0$, and $v^2 \notin E$ by Theorem 1.6,

$$0 \neq [v^2] \in \ker \psi_v, \text{quad rank } \psi_v \leq 99. \quad (7)$$

Proposition 1.8. *For every $0 \neq v \in V$,*

$$\text{rank } \psi_v = 99, \text{quad } \ker \psi_v = \text{span}([v^2]).$$

Proof. First take the affine chart $v_{00} \neq 0$. Use the lexicographic bases

$$\text{Sym}^2 V : x_a x_b \ (0 \leq a \leq b < 16), \quad V \otimes \bigwedge^2 V : x_a \otimes (x_b \wedge x_c) \ (b < c).$$

The program selects 560 independent columns of $\delta(E \otimes V)$, their 560 pivot rows, 99 classes in $\text{Sym}^2 V/E$, and 99 additional rows. For $v = \sum_{k=0}^{15} v_k x_k$, with $v_0 = v_{00}$, the resulting minor is

$$\mathcal{M}(v) = \begin{bmatrix} A_0 & Q_0(v) \\ A_1 & Q_1(v) \end{bmatrix}, \quad (8)$$

where A_0 is 560×560 and the right side has 99 columns. Exact rational elimination gives $\det A_0 = 1$. Put

$$S(v) := Q_1(v) - A_1 A_0^{-1} Q_0(v) = \sum_{k=0}^{15} v_k S_k.$$

Direct calculation over $\mathbb{Q}$ gives

$$\det S_0 = -32768. \quad (9)$$

For $k = 1, \dots, 15$, set $N_k = S_0^{-1} S_k$. The exact output verifies that every N_k has zero diagonal, that their combined support has 156 positions, and that the directed graph formed by these positions is acyclic. A common topological ordering therefore makes all N_k strictly upper triangular. Hence

$$\begin{aligned} \det \mathcal{M}(v) &= \det A_0 \det S_0 \det \left(v_0 I_{99} + \sum_{k=1}^{15} v_k N_k \right) \\ &= -32768 v_{00}^{99}. \end{aligned} \quad (10)$$

This is nonzero when $v_{00} \neq 0$, so $\text{rank } \psi_v \geq 99$. Equation (7) gives equality and identifies the kernel.

The group $S_4 \times S_4$ permutes the matrix rows and columns, preserves perm_4 , E , and δ , and acts transitively on the sixteen coordinates. Every nonzero v has a nonzero coordinate, which can be moved to $(0, 0)$. This reduces the general case to the chart already proved. $\square$

Remark 1.9. Equation (10) is an identity over $\mathbb{Q}$, not a numerical fit. Modular computation only selects candidate row and column indices; the determinant and common triangularization are reconstructed exactly over $\mathbb{Q}$.

1.5. Adding one quadratic direction.

Lemma 1.10. *If $q \in \text{Sym}^2 V \setminus E$, then*

$$\text{rank } \delta((E + \text{span}(q)) \otimes V) \geq 575.$$

Proof. Define the linear map

$$\theta_q : V \longrightarrow \text{coker}(\delta(E \otimes V)), \quad v \longmapsto \psi_v([q]).$$

Suppose independent v, w lie in $\ker \theta_q$. Since $[q] \neq 0$, Proposition 1.8 gives nonzero scalars a, b such that $[q] = a[v^2] = b[w^2]$ in $\text{Sym}^2 V/E$. Thus

$$av^2 - bw^2 \in E.$$

The left side is nonzero and has rank at most two, contradicting Theorem 1.6. Hence $\dim \ker \theta_q \leq 1$ and $\text{rank } \theta_q \geq 15$. Its image is exactly the new image contributed by $q \otimes V$, so

$$\text{rank } \delta((E + \text{span}(q)) \otimes V) \geq 560 + 15 = 575. \quad \square$$

1.6. A rank inequality.

Lemma 1.11. *For two $m \times n$ matrices A, B ,*

$$\text{rank}(A - B) \geq \text{rank} \begin{bmatrix} A & B \end{bmatrix} + \text{rank} \begin{bmatrix} A \\ B \end{bmatrix} - \text{rank } A - \text{rank } B. \quad (11)$$

Proof. Let

$$c = \dim(\text{im } A \cap \text{im } B), \quad d = \dim(\text{row } A \cap \text{row } B).$$

Then the horizontal and vertical concatenations have ranks $\text{rank } A + \text{rank } B - c$ and $\text{rank } A + \text{rank } B - d$, respectively. For $x \in \ker(A - B)$, the vector $Ax = Bx$ lies in $\text{im } A \cap \text{im } B$, and the kernel of $x \mapsto Ax$ on $\ker(A - B)$ is $\ker A \cap \ker B$. Therefore

$$\dim \ker(A - B) \leq c + n - \text{rank } A - \text{rank } B + d.$$

Taking codimension in n gives (11). $\square$

Remark 1.12. Replacing B by αB , $\alpha \neq 0$, does not change either concatenation rank. Thus (11) also controls $A - \alpha B$.

1.7. Excluding seven terms.

Theorem 1.13.

$$\text{ChowRank}(\text{perm}_4) \geq 8.$$

Proof. Fix a nonzero Chow term T , and set

$$A = K(\text{perm}_4), \quad B = K(T), \quad F = E_T, \text{quadd} = \dim(E \cap F).$$

Corollary 1.7 gives $d \in \{0, 1\}$.

If $d = 0$, then (3) gives

$$\text{row } A \subseteq E \otimes V^*, \quad \text{row } B \subseteq F \otimes V^*,$$

and these ambient spaces intersect trivially. Thus the vertical concatenation has rank $\text{rank } A + \text{rank } B$. Lemma 1.11, together with $\text{rank}[A \ B] \geq \text{rank } A = 560$, gives

$$\text{rank}(A - \alpha B) \geq 560 \quad (\alpha \neq 0). \quad (12)$$

If $d = 1$, choose $0 \neq q \in E \cap F$. Since $F \subseteq \text{Sym}^2 \text{span}(\ell_1, \dots, \ell_4)$ and q has rank four, the factors span a four-dimensional space and are independent. Hence $\dim F = 6$ and $\text{rank } B = 92$. Choose $r \in F \setminus E$. The image of the horizontal concatenation contains $\delta((E + \text{span}(r)) \otimes V)$, so Lemma 1.10 gives

$$\text{rank}[A \ B] \geq 575. \quad (13)$$

Also,

$$\dim(\text{row } A \cap \text{row } B) \leq \dim((E \cap F) \otimes V^*) = 16,$$

and hence

$$\text{rank} \begin{bmatrix} A \\ B \end{bmatrix} \geq 560 + 92 - 16 = 636. \quad (14)$$

Lemma 1.11 now gives

$$\text{rank}(A - \alpha B) \geq 575 + 636 - 560 - 92 = 559 \quad (\alpha \neq 0). \quad (15)$$

Thus subtracting any nonzero Chow term from perm_4 leaves a flattening of rank at least 559. If perm_4 had a seven-term decomposition, remove one nonzero term. The remaining six terms would have flattening rank at most $6 \cdot 92 = 552$, contradicting (12) or (15). A decomposition with fewer nonzero terms gives the same contradiction. Therefore $\text{ChowRank}(\text{perm}_4) \geq 8$. $\square$

1.8. The eight-term decomposition. Glynn's identity is

$$\text{perm}_n = 2^{1-n} \sum_{\substack{\varepsilon \in \{\pm 1\}^n \\ \varepsilon_0 = 1}} \left(\prod_{i=0}^{n-1} \varepsilon_i \right) \prod_{j=0}^{n-1} \left(\sum_{i=0}^{n-1} \varepsilon_i x_{ij} \right). \quad (16)$$

The sum has 2^{n-1} terms, each a product of n linear forms. For $n = 4$, it gives $\text{ChowRank}(\text{perm}_4) \leq 8$. Together with Theorem 1.13, this proves Theorem 1.2.

1.9. Exact computation. The computational part has a primary implementation and an independent reconstruction.

File	Purpose
<code>perm4_quadratic_extension_chart_exact_verify.py</code>	Reconstructs (8) over $\mathbb{Q}$, computes the Schur complement and $\det S_0 = -32768$, and checks simultaneous triangularization.
<code>perm4_quadratic_extension_full_minor_replay.py</code>	Replays the full 659×659 minor at several parameter points over two primes as a separate cross-check.
<code>perm4_rank8_independent_audit.py</code>	Rebuilds the ranks 560 and 92, the cubic prolongations, and the chart minor directly from the displayed bases without importing project modules.
<code>perm4_rank8_verify_all.py</code>	Runs the upper bound, base ranks, chart calculation, and full-minor replay.

The independent program reports

```
independent_perm3_prolongations=16
independent_chow_koszul_rank=92
independent_chow_triple_prolongations=4
direct_E_image_rank=560
independent_chart_constant=-32768
independent_chart_triangular=True
INDEPENDENT_RANK8_AUDIT_PASS
```

The lower bounds 560 and 92 come from displayed integer minors that are nonzero modulo 1000003; the matching upper bounds come from the sixteen and four explicit cubic prolongations. The identity $\det \mathcal{M}(v) = -32768v_{00}^{99}$ is verified over $\mathbb{Q}$, and $S_4 \times S_4$ symmetry carries that chart to every nonzero v .

From the `perm4/certificates` directory, run

```
python perm4_rank8_independent_audit.py
python perm4_rank8_verify_all.py
```

Both commands must exit successfully; the first prints `INDEPENDENT_RANK8_AUDIT_PASS`, and the second prints its aggregate success marker.

The lower bound above is proved over the complex numbers. Any decomposition over a characteristic-zero field remains a decomposition after scalar extension to the complex numbers, while Glynn's decomposition is defined over the prime field. The equality therefore holds over every field of characteristic zero.